1 Orthogonal Trajectories

An *orthogonal trajectory* of a family of curves is a curve that intersects each curve of the family orthogonally, that is, at right angles.

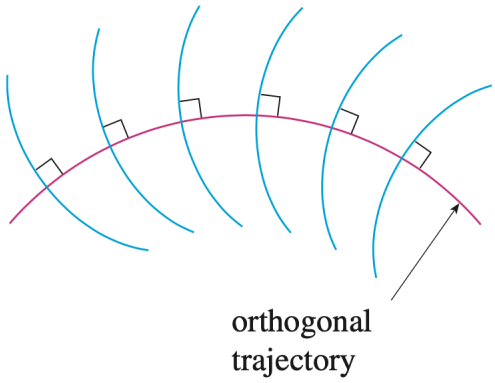


Figure 1: An orthogonal trajectory.

When we say that two curves intersect at a right angle, we mean that their tangent lines are perpendicular at the intersection point.

Example Each member of the family $y = mx$ of straight lines through the origin is an orthogonal trajectory of the family $x^2 + y^2 = r^2$ of concentric circles centered at the origin. We say that the two families are orthogonal trajectories of each other.

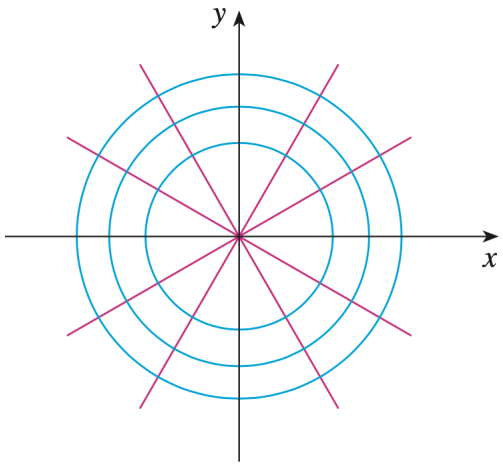


Figure 2: Two families of curves that are orthogonal trajectories of each other.

1. We illustrate the method of finding orthogonal trajectories with the following example. Consider the family of curves $x = ky^2$, where k is an arbitrary real number.

   (a) Differentiate both sides of the equation $x = ky^2$ with respect to x and solve for $\frac{dy}{dx}$, the slope of the tangent line to the curve determined by k.

   (b) Use the equation $x = ky^2$ to eliminate k from your expression for $\frac{dy}{dx}$, so that it only depends on x and y.

   (c) Your answer to part (b) gives the slope of the tangent line to a given curve in the family at the point (x,y). Find the slope $m(x,y)$ of an orthogonal line at the point (x,y).

   (d) Solve the differential equation $y' = m(x,y)$ to find the family of orthogonal trajectories. Plot some of the curves in each family in Desmos to verify that these families are indeed orthogonal trajectories of each other.

2 Exercises

Find the orthogonal trajectories of each family of curves. Use Desmos to plot several members of each family.

2. $x^2 + 2y^2 = k^2$

3. $y^2 = kx^3$

4. $y = \dfrac{k}{x}$

5. $y = \dfrac{x}{1+kx}$